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PhcQ mainly contributes to the regulation of quorum sensing-dependent genes, in which PhcR is partially involved, in *Ralstonia pseudosolanacearum* strain OE1-1

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Abstract

The gram-negative plant-pathogenic β-proteobacterium *Ralstonia pseudosolanacearum* strain OE1-1 produces methyl 3-hydroxymyristate as a quorum sensing (QS) signal via the methyltransferase PhcB and senses the chemical through the sensor histidine kinase PhcS. This leads to functionalization of the LysR family transcriptional regulator PhcA, regulating QS-dependent genes responsible for the QS-dependent phenotypes including virulence. The *phc* operon consists of *phcB*, *phcS*, *phcR*, and *phcQ*, with the latter two encoding regulator proteins with a receiver domain and a histidine kinase domain and with a receiver domain, respectively. To elucidate the function of PhcR and PhcQ in the regulation of QS-dependent genes, we generated *phcR*-deletion and *phcQ*-deletion mutants. Though the QS-dependent phenotypes of the *phcR*-deletion mutant were largely unchanged, deletion of *phcQ* led to a significant change in the QS-dependent phenotypes. Transcriptome analysis coupled with quantitative reverse transcription-PCR and RNA-sequencing revealed that *phcB*, *phcK*, and *phcA* in the *phcR*-deletion and *phcQ*-deletion mutants were expressed at similar levels as in strain OE1-1. Compared with strain OE1-1, expression of 22.9% and 26.4% of positively and negatively QS-dependent genes, respectively, was significantly altered in the *phcR*-deletion mutant. However, expression of 96.8% and 66.9% of positively and negatively QS-dependent genes, respectively, was significantly altered in the *phcQ*-deletion mutant. Furthermore, a strong positive correlation of expression of these QS-dependent genes was observed between the *phcQ*-deletion and *phcA*-deletion mutants. Our results indicate that PhcQ mainly contributes to the regulation of QS-dependent genes, in which PhcR is partially involved.

Chika Takemura and Wakana Senuma contributed equally to this work.

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